Set A

جامعة التقنية والعلوم التطبيقية
University of Technology and Applied Sciences

نزوى
Nizwa

College of Computing and Information Sciences
Department of Information Technology

Final Examination
Fall, AY 2023/2024
CSPG1205 – Introduction to Programming

Date: Sunday, January 14, 2024		**Time:** 09:00 AM – 11:00 AM	
Max Marks	40	Time Allocation	2 Hour

Student ID		Student Name	
Section		Level	Diploma

Instructions to the Students

1. Answer the questions in the space provided by using **blue or black pen** only. You will obtain a zero (0) if you use a pencil for all question.
2. This exam paper consists of **(7)** pages including the front page.
3. Use of mobile phones and any electronic gadgets inside the examination hall are not allowed.
4. ***Copying, Cheating, and any kind of malpractices on the examination are strictly prohibited.*** Action will be taken as per the "Article 68 of UTAS Academic Regulations".
5. Students should bring their own pencils, pens, erasers, rulers, calculators, and any other tools required for the exam.
6. The exam has a total of 14 **non optional questions**.
7. Use of calculator or any other electronic device is not allowed.

Section	Allotted Marks	Obtained Marks
A	10	
B	15	
C	15	
Total Marks	40	

First Marker	Second Marker

Section A: Choose the best answer 10 marks

Q1. Which of the following is **true** about a compiler?

a) It is a program that is used to translate high-level language program into machine code.

b) It is a program that translates assembly language program to machine code.

c) It is an application program to edit and debug Python program.

d) It is a hardware component that execute Python program.

Q2. Which one of the following statements produces the following output?

```
I'm fine.
```

a) `print ('I'm fine.')`

b) `print ('I\'m fine.')`

c) `print ('I/m fine.')`

d) `print ("I\\'m fine.")`

Q3. What is the output of the following code?

```
a = 1
while a < 10:
    a = a + 1
    if a % 2 == 0:
        continue
    print (a,end=" ")
    if a % 5 == 0:
        break
```

a) 1 3 5 7 9

b) 3 5

c) 1 3 5

d) 3 5 7 9

Q4. Which one of the following Python codes prints the first three odd numbers (1, 3, 5)?

a)
```
for i in range (1,5):
    print (i)
```

b)
```
for i in range (1,6):
    print (i)
```

c)
```
for i in range (1,5,2):
    print (i)
```

d)
```
for i in range (1,6,2):
    print (i)
```

Q5. What is the output of the following code?

```
def f1(x,y=10):
    print (x+y, end = ", ")
def f2(x=10):
    print (x, end = ", ")
    f1(x)
    return x+2

print (f2(5))
```

a) 5, 15, 7

b) 15, 7, 5

c) 5, 7, 15

d) 15, 5, 7

Q6. Which one of the following statements is used to find the number of times a substring `s2` is present in a string `s1`?

a) `s1.find (s2)`

b) `s2.find (s1)`

c) `s2.count (s1)`

d) `s1.count(s2)`

Q7. Consider the following string

s = “University of Technology and Applied Sciences-Nizwa”

Which one of the following statements is used to replace Nizwa with Muscat.

a) `s = replace (“Nizwa”,”Muscat)`

b) `s = replace (“Nizwa”,”Muscat”)`

c) `s = s.replace(“Nizwa”,”Muscat”)`

d) `s = s.replace(“Muscat”,”Nizwa”)`

Q8. Which one of the following statements is used to read ***n*** characters from a file?

a) `s = fp.readlines(n)`

b) `n = fp.read()`

c) `s = fp.read(n)`

d) `s = fp.readlines(n)`

Q9. Which one of the following statements is ***false*** about modules?

a) Modules can’t import other modules

b) Modules can be reused

c) Modules can be imported by other programs to use its functionality

d) Module is a file containing a set of functions and other statements.

Q10. Which one of the following is not the name of a standard module in Python?

a) `math`

b) `re`

c) `os`

d) `findall`

Section A: ___/10

Section B:

Q11. Correct the errors in the following recursive and non-recursive functions to find the factorial of a number and rewrite these functions without errors in the space given below.

8 marks

```
def factorial (n):
    fact = 0
    for i in range (n,0,1) :
        fact =* i
    return factorial
```

```
def factorial (n) :
    fact = 1
    for i in range (n,0,-1) :
        fact *= i
    return fact
```

```
def factorial (n):
    if n = 0 :
        return 0
    else
        return n * fact(n-1)
```

```
def factorial (n) :
    if n == 0 :
        return 1
    else :
        return n * factorial(n-1)
```

8 errors: Each error 1 mark.

Q12. The following code with some missing parts accepts the marks of ***n*** students and calculate the grades according to the following table. The program creates two lists to store marks and grades and displays the result as a table. 7 marks

Mark	Grade
0-49	F
50-70	B
71-100	A

Choose the words from the following table to fill the missing parts

and	elif	int	mark	append
else	if	or	m_list[i]	float
insert	add	g_list[i]	grade	

```
n = int (input ("Enter number of students:"))
m_list = []
g_list = []
for i in range (n):
    mark = int (input ("Enter marks:"))
    if mark >= 0 and mark <= 49:
        grade = 'F'
    elif mark >=50 and mark <= 70:
        grade = 'B'
    else :
        grade = 'A'
    m_list. append (mark)
    g_list. append (grade)
print ("\nMark\tGrade")
for i in range (n):
    print (m_list[i],"\t\t",g_list[i])
```

7 blanks: Each 1 mark.

Section B: ___/15

Section C:

Q13: Write a program to make a copy of a text file. Handle possible file exceptions.

Accept the names of the files from the user as input.

7 marks

```
fname1=input ("Enter source file name:")
fname2=input ("Enter destination file name:")


try :
    fp1=open (fname1,"r")
    fp2=open (fname2,"w")
    s = fp1.read ()
    fp2.write (s)
    fp1.close()
    fp2.close()
except FileNotFoundError:
    print ("Source file not found")
except PermissionError :
    print ("Can't write. Permission denied")
```

Accepting file names :1 mark
Opening files : 1 mark
Read : 1 mark
Write : 1 mark
Close : 1 mark
Exception handling : 2 mark
(One try/except block is also OK)

Q14. Write a program to accept the name and age of ***n*** students and store them in a dictionary with name as key and age as value.

The program should access the names from this dictionary and use regular expression to print all names that start with a vowel.

8 marks

```
import re
d1 = {}
n = int (input ("enter n :"))
for i in range (n) :
    name = input ("Enter name :")
    age = int (input ("Enter age :"))
    d1[name] = age
for i in d1.keys() :
    if re.findall(r"\b[aeiouAEIOU]\w+\b",i):
        print (i)
```

Import re :0.5 mark
Defining empty dictionary :0.5 mark
input : 1 mark
Adding to dictionary :1 mark
for loop to get the keys: 1 mark
if and Regular expression: 3 marks
print :1

Section C: ___ / 15